Chapter 1: History of Computing

Define computing. Explain all the major innovations in the field of hardware, software and networking. OR some of the major breakthroughs.

Computing is the process of using computers and computing devices to perform various tasks and solve problems. It involves inputting data, processing it, and producing output or results.

Major Innovations in the Field of Computing:

Hardware Innovations:

- **Integrated Circuits**: The invention of integrated circuits (ICs) led to the decrease of electronic components, enabling the creation of smaller, faster, and more powerful computers.
- **Microprocessors**: Microprocessors are single-chip CPUs that revolutionized computing by bringing the processing power of a whole computer onto a single chip, making personal computers and mobile devices possible.
- **Personal Computers**: The development of personal computers (PCs) made computing accessible to individuals, leading to widespread adoption and use in homes and offices.
- **Graphical User Interface (GUI)**: GUIs introduced visual icons, windows, and menus, making computers more user-friendly and allowing users to interact with computers using a mouse instead of complex commands.

Software Innovations:

- **Operating Systems**: The creation of operating systems like MS-DOS, Windows, macOS, and Linux provided the foundation for managing hardware resources and running applications.
- **Graphical User Interface (GUI)**: GUIs revolutionized software interaction, making it easier for users to navigate and interact with applications using visual elements.
- **Internet Browsers**: Web browsers like Netscape Navigator and Internet Explorer made it possible to browse the World Wide Web, unleashing the potential of the internet for communication and information access.
- **Open-Source Software**: The concept of open-source software, where the source code is freely available for modification and distribution, has led to a vast ecosystem of collaborative software development.

Networking Innovations:

- **ARPANET**: The development of ARPANET in the 1960s laid the foundation for the internet, connecting computers and allowing them to communicate and share information.
- **World Wide Web (WWW)**: The invention of the World Wide Web by Tim Berners-Lee made it easy to navigate and access information on the internet using hyperlinks, giving rise to modern web browsing.
- **TCP/IP Protocol**: The TCP/IP protocol suite standardized communication between devices on the internet, ensuring seamless data transmission across diverse networks.
- **Wireless Networks**: Wireless technologies like Wi-Fi and cellular networks enabled mobile computing, allowing users to access the internet and communicate on the go.

Discuss the contribution of Conrad Zuse in the field of computing.

Conrad Zuse, a German engineer, and inventor, made significant contributions to the field of computing.

Major Contributions of Conrad Zuse:

- **Z1 Computer (1938-1939)**: Conrad Zuse designed and built the Z1, which is considered the world's first programmable, fully automatic, binary digital computer. It used mechanical relays for calculations and had a program stored on punched film.
- **Z3 Computer (1941-1945)**: Zuse's most significant contribution was the Z3 computer, completed in 1941. The Z3 was the first working programmable, fully automatic, electromechanical computer. It was the world's first program-controlled computer to use binary floating-point arithmetic. The Z3 could solve complex engineering and scientific problems.
- **Plankalkül (1945-1946)**: Conrad Zuse developed the Plankalkül, one of the earliest high-level programming languages. Although not implemented during Zuse's time, Plankalkül introduced several fundamental concepts, such as variables, loops, and arrays, which laid the foundation for modern programming languages.
- **Z4 Computer (1942-1945)**: During World War II, Zuse developed the Z4, an improved version of the Z3, which used binary floating-point representation and faster relay technology. The Z4 was used for scientific and engineering calculations after the war.
- **S1 Computer (1950)**: After the war, Zuse started a company to manufacture computers. He built the Z4 for ETH Zurich and later developed the S1, a transistorized computer. The S1 was one of the earliest computers to use vacuum tubes and transistors.

Describe how the computing changes with the evolution in computer technology?

Miniaturization and Portability: Early computers were massive, but advances in semiconductor technology led to miniaturization, making computers smaller and more portable. This evolution gave rise to laptops, smartphones, and tablets, enabling computing on the go.

Processing Power: Moore's Law, which states that the number of transistors on a microchip double approximately every two years, has driven rapid increases in processing power. Modern computers are much faster and more powerful.

Memory and Storage: Early computers had limited memory and storage capacities. Advancements in memory technology, such as RAM and solid-state drives (SSDs), have increased computer memory and storage capabilities, allowing for the handling of large datasets and faster data access.

Graphical User Interface (GUI): The development of GUIs made computers more user-friendly, allowing users to interact with computers using visual elements like icons, windows, and menus.

Networking and the Internet: The evolution of computer networking and the development of the internet revolutionized how computers communicate and share information. The internet connects computers worldwide, facilitating global communication and access to vast amounts of information.

Cloud Computing: Cloud computing has transformed how computing resources are accessed and utilized. It allows users to store and access data and run applications over the internet, reducing the need for local infrastructure and enabling scalable computing power.

Describe the contribution of Edward Djkstra in the field of computing.

Shortest Path Algorithm: Dijkstra is best known for his algorithm for finding the shortest path in a graph. The Dijkstra's algorithm is widely used in computer networks, transportation systems, and various optimization problems.

Structured Programming: Dijkstra was a strong advocate for structured programming, which emphasizes the use of structured control flow constructs like loops and conditionals to improve the clarity and maintainability of code.

Semaphore: Dijkstra introduced the concept of semaphores, a synchronization primitive used for managing concurrent access to shared resources in concurrent computing.

Deadlock Prevention: Dijkstra's work on deadlock prevention and avoidance in concurrent systems significantly contributed to the understanding and mitigation of deadlocks, a common problem in multi-threaded environments.

Structured Text Editor (Goto-less programming environment): Dijkstra designed a programming environment known as the "Structured Text Editor," which encouraged structured programming by disallowing the use of the "goto" statement.

Difference between all generations of computer.

First Generation (1940s-1950s):

- **Technology**: Used vacuum tubes as the main electronic component.
- **Size and Cost**: Large, expensive, and required a significant amount of power and cooling.
- **Programming**: Programmed using machine language and low-level assembly languages.
- **Speed and Memory**: Slow processing speed and limited memory capacity (measured in kilobytes).
- **Examples**: ENIAC, UNIVAC, EDVAC.

Second Generation (1950s-1960s):

- **Technology**: Replaced vacuum tubes with transistors, making computers smaller, more reliable, and energy efficient.
- **Programming**: Still used low-level assembly languages but saw the development of high-level programming languages like FORTRAN and COBOL.
- **Speed and Memory**: Faster processing speeds and improved memory capacity (measured in kilobytes to megabytes).
- **Examples**: IBM 1401, IBM 7090, UNIVAC II.

Third Generation (1960s-1970s):

- **Technology**: Introduced integrated circuits (ICs), with multiple transistors on a single chip, further reducing computer size and power consumption.
- **Programming**: High-level programming languages became more prevalent, improving software development productivity.
- **Speed and Memory**: Faster processing speeds and increased memory capacity (measured in megabytes).
- **Examples**: IBM System/360, DEC PDP-8, CDC 6600.

Fourth Generation (1970s-1980s):

- **Technology**: Utilized large-scale integration (LSI) and very large-scale integration (VLSI) chips, allowing for even smaller and more powerful computers.
- **Programming**: Higher-level programming languages, structured programming, and object-oriented programming gained prominence.
- **Speed and Memory**: Much faster processing speeds and expanded memory capacity (measured in megabytes to gigabytes).

- **Examples**: IBM System/370, DEC VAX, Apple II, IBM PC.

Fifth Generation (1980s-Present):

- **Technology**: Introduced microprocessors and microcontrollers, enabling computers to be embedded in various devices and appliances.
- **Programming**: Continued advancements in high-level languages, object-oriented programming, and the development of scripting languages.
- **Speed and Memory**: Dramatically faster processing speeds and significantly increased memory capacity (measured in gigabytes to terabytes).
- **Examples**: IBM PC AT, Macintosh, modern desktops, laptops, smartphones, and embedded systems.

Chapter 2: Social Context of Computing

1. Impact of Technology on Society.

Positive Impact:

a. Improved Communication

Technology had made it easier for people to communicate with each other, regardless of distance. This has enabled people to stay connected with family, friends and access information from anywhere in the world.

b. Increased Efficiency

Many technologies have improved the efficiency of various industries, making it easier to produce goods and services at a faster pace. This has helped to increase productivity and reduce costs.

c. Improved healthcare

Technology has improved the quality of healthcare services, making it easier for doctors and healthcare providers to diagnose and treat patients. Medical machines such as MRI machines, CT scanners, robotic surgery has revolutionized the way we treat diseases and illness.

d. Access to information

The internet has made it easier for people to access information from anywhere in the world. This has enabled people to learn new things, stay informed about currents events and engage with other people with similar interest.

e. Increased Convenience: E-commerce and online services have made it easier for people to shop, bank, and access various services from the comfort of their homes, saving time and effort.
f. Online Communities: Computers have given rise to online communities and forums where like-minded individuals can connect, share interests, and support each other, fostering a sense of belonging.
g. Cloud computing ko barema lekhnu

Negative Impact

a. Social Isolation

While technology has made it easier for people to communicate with each other easily, it has also led social isolation for some individuals. Spending too much time online can lead to lack of face-to-face interactions and reduced social skills.

b. Job Loss

Many technological advancements, such as automation and AI, have the potential to replace human workers in certain industries. This can lead to job loss and economic inequality.

c. Cybersecurity threats

As technology becomes more integrated into our lives, the risk of cyber-attack and data breaches increases. This can lead to loss of sensitive information and a loss of trust from technology.

d. Addiction

Technology addiction or excessive use of technology can have a negative impact on mental health and wellbeing. It can lead to feelings of anxiety, depression etc.

2. Explain health related issues for an IT professional.

- **Computer Vision Syndrome (CVS)**: Long screen time can lead to eye strain, dry eyes, headaches, and blurred vision. Implementing the 20-20-20 rule (looking at something 20 feet away for 20 seconds every 20 minutes) and using proper lighting can help alleviate CVS symptoms.
- **Sleep Disruptions**: Excessive screen time, especially before bedtime, can disrupt sleep patterns and lead to sleep disturbances, insomnia, and fatigue.
- **Healthcare Neglect**: Due to busy work schedules, some IT professionals may neglect regular health check-ups and fail to seek medical attention for minor health issues, leading to potential health complications.
- **Isolation and Social Interaction**: IT professionals may spend long hours working alone or in isolated environments, which can lead to reduced social interaction and feelings of loneliness.
- **Unhealthy Eating Habits**: Long working hours and stressful environments can lead to irregular eating habits and reliance on fast food or unhealthy snacks, contributing to poor nutrition and weight gain.
- **Stress and Mental Health**: IT professionals often work under tight deadlines and deal with complex problem-solving tasks, leading to high levels of stress and pressure. This can impact mental health, causing anxiety, depression, and burnout.
- **Musculoskeletal Problems**: Poor ergonomics and prolonged sitting can cause musculoskeletal issues, including neck pain, back pain, shoulder pain, and repetitive strain injuries (RSIs) like carpal tunnel syndrome.

3. What do you mean by society? Describe the essentials elements of society. Or features of Society.

Society is a complex and organized group of individuals living together in a particular geographical area and sharing common customs, values, and institutions.

Essential Elements of Society:

- **Plurality**: Society is composed of diverse individuals who possess unique characteristics, perspectives, and backgrounds. Plurality recognizes the existence of different social groups, ethnicities, religions, and cultures within the larger social fabric.
- **Stability**: A stable society is one where there is order and balance, with established rules, norms, and institutions that ensure peaceful coexistence and prevent chaos.
- **Likeness**: While society embraces diversity, there are shared values, traditions, and identities that create a sense of unity and cohesion (sticking together) among its members.
- **Differences**: Despite likeness, individuals in society have unique opinions, abilities, and characteristics, adding depth and complexity to social interactions.
- **Interdependence**: Society functions effectively because people rely on each other for various needs, resources, and support, recognizing that cooperation and collaboration are essential for mutual progress.
- **Cooperation**: Cooperation is vital for the smooth functioning of society. People work together, helping each other and achieving common goals, leading to collective prosperity and well-being.

4. **What do you understand by 'Social Context of Computing'?**

In simple words, the "Social Context of Computing" refers to how computers and technology are influenced by the society in which they are used and, in turn, how they impact people and society.

5. **State and explain any FOUR issues that are a major concern to the Society and Internet.**

Privacy and Data Security: As more of our lives are conducted online, concerns about privacy and data security have become paramount. Users worry about their personal information falling into the wrong hands or being misused for targeted advertising or other purposes. Companies and governments must implement robust data protection measures and strict to strict privacy regulations to address these concerns.

Digital Divide and Access Inequality: The digital divide refers to the gap between those who have access to digital technologies and those who do not. Lack of internet access and digital literacy can hinder individuals' educational and economic opportunities. Governments and organizations need to invest in infrastructure and initiatives to bridge this gap, ensuring that all members of society can benefit from the opportunities the internet offers.

Misinformation and Fake News: The internet has enabled the rapid dissemination of information, but it has also given rise to the spread of misinformation and fake news. False information can lead to public confusion, social unrest, and the erosion of trust in traditional media and institutions. Addressing this issue requires promoting media literacy and responsible information-sharing practices.

6. Explain about digital divide with example.

The digital divide refers to the gap between those who have access to modern information and communication technologies (ICTs) like the internet and computers, and those who do not.

Example: Access to Internet and Education

In a rural village in a developing country, there is limited access to the internet and modern technology. The village lacks reliable electricity and internet infrastructure, making it difficult for residents to connect to the digital world. As a result, the students attending the local school have limited access to educational resources available online. They do not have access to e-learning platforms, online research materials, or educational apps that could enhance their learning experience.

On the other hand, in a city in a developed country, students attending a well-funded school have access to a fully equipped computer lab and high-speed internet. They can access a wide range of educational resources, take online courses, collaborate with students from other countries, and stay up to date with the latest developments in their fields.

In this scenario, the digital divide creates unfair in educational opportunities. The students in the rural village are at a disadvantage compared to their urban counterparts, as they lack access to the same digital tools and resources that could enrich their learning experiences and better prepare them for the future.

The digital divide is not limited to education; it also affects access to information, job opportunities, healthcare services, and social connectivity.

7. Explain about governance of internet with example.

The governance of the internet refers to the processes and mechanisms used to manage and regulate its various aspects, including technical standards, domain names, cybersecurity, content regulation, and privacy. The internet is a global network that connects billions of users and devices, and its governance involves multiple stakeholders, including governments, private organizations, technical experts, and civil society. Internet governance is crucial to ensure the internet's stability, security, and accessibility for all users worldwide. Here's an example to illustrate internet governance:

ICANN is a non-profit organization responsible for the coordination of the global domain name system (DNS) and IP address allocation. It plays a central role in the governance of the internet's technical infrastructure. The DNS translates human-readable domain names (like example.com) into machine-readable IP addresses (like 192.0.2.1), allowing users to access websites and services on the internet.

For instance, if a company wants to register a new top-level domain (TLD) like .example, they must apply to ICANN following specific guidelines and pay a fee. ICANN evaluates the application, considering technical, operational, and public interest aspects. The decision-making process involves public comments, evaluations by experts, and potential objections from stakeholders.

8. **What are the social implications of Information Technology/Network? Explain in detail. OR**

Identify and explain issues about technology and social change that often underline discussions of the social implications of computer and information technology. (Write only negative parts)

Communication and Connectivity:

- Positive: IT has revolutionized communication, enabling instant and global connectivity through email, social media, and messaging apps. People can now easily connect with friends, family, and colleagues worldwide, fostering social bonds and cultural exchange.
- Negative: Overreliance on digital communication can lead to reduced face-to-face interactions, potentially causing social isolation and a decline in the quality of personal relationships.

Education and Knowledge Sharing:

- Positive: IT has democratized access to information and education. Online learning platforms, educational apps, and digital libraries provide learning opportunities to a vast number of people, irrespective of their location or socioeconomic status.
- Negative: The availability of vast amounts of information can lead to information overload, making it challenging to discern accurate and reliable sources from misleading or false content.

Economic Impact and Employment:

- Positive: IT has created new job opportunities in the technology sector, contributing to economic growth and innovation. E-commerce and online marketplaces have expanded business opportunities for entrepreneurs and small businesses.

- Negative: Automation and artificial intelligence may lead to job displacement in certain industries, creating concerns about unemployment and income inequality.

Privacy and Data Security:

- Positive: IT has facilitated seamless data exchange and personalized services. Data analytics and machine learning help improve user experiences and optimize services.
- Negative: The collection and use of personal data raise privacy concerns. Data breaches and cyber-attacks can lead to identity theft and misuse of sensitive information, threatening individuals' privacy and security.

Or question ko further answer
Digital Divide: The digital divide refers to the gap between those who have access to technology and those who do not. The uneven distribution of technology can exacerbate existing social inequalities, hindering marginalized communities' access to education, job opportunities, and essential services.
Disinformation and Fake News: The rapid spread of false information and misinformation on social media and the internet can lead to public confusion and undermine trust in reliable sources. This issue has significant implications for democratic processes and public discourse.

9. **How can we minimize Digital Divide?**

Government Needs to Step Up:

Governments can significantly reduce the digital divide gap if they reduce the prices of technological devices, internet services, electricity taxes.

In the future, policymakers and educational leaders should connect with governments and big corporations to help schools, learners, and teachers access digital technologies. This collaboration will reduce the cost of the internet and strengthen the technological infrastructure.

More Digital Learning

Schools need to also step up with the governments and introduce technical learning curriculums in the classrooms. In addition, teachers and students need to be fully trained in using digital resources to uplift the education standards.

Schools need to host seminars and workshops to provide technical training to teachers.

Use Local Languages in Online Learning Materials

Online learning resource creators should focus on including local languages in their programs. According to World Bank research, people feel more connected to content when they read it in their local language. Moreover, many people worldwide are not fluent in the English language.

Public Access Centers:

- Establish public access centers, such as libraries, community centers, or schools, with free internet and computer access.
- These centers can serve as hubs for digital learning and communication for those without personal devices or connectivity.

Infrastructure Development:

- Improve internet infrastructure in underserved areas by expanding broadband connectivity and mobile networks.
- Invest in rural and remote communities to ensure they have access to high-speed internet services.

10. **Information Technology has a major impact on poverty alleviation. Do you agree or disagree? Justify with your opinion.**

- **Access to Information and Education**:

IT enables access to a wealth of information and educational resources. With the internet, individuals in impoverished regions can learn new skills, access online courses, and acquire knowledge that can improve their employability and economic prospects.

- **Economic Empowerment**:

IT has transformed the job market, creating remote work opportunities, freelance platforms, and online marketplaces. This allows individuals from impoverished backgrounds to participate in the global economy and earn a livelihood.

- **Financial Inclusion**:

IT-based financial services, such as mobile banking and digital payment systems, have expanded financial inclusion. People in poverty can access formal financial services, save money, and access credit more easily.

- **E-Commerce and Entrepreneurship**:

IT has facilitated e-commerce, enabling small businesses and entrepreneurs to reach a broader customer base. This can lead to increased sales and revenue, contributing to poverty reduction.

- **Agricultural Advancements**:

IT solutions, such as precision farming and weather forecasting apps, have improved agricultural productivity. Farmers can make informed decisions, optimize resources, and increase their income.

- **Healthcare and Telemedicine**:

IT has enhanced healthcare access in remote and underserved areas through telemedicine and mobile health apps. People can receive medical consultations and health information remotely, improving health outcomes.

- **Education and Digital Literacy**:

IT enables digital literacy programs, improving technology skills and digital understanding. This enhances individuals' employability and their ability to navigate the digital world effectively.

11. Define E-governance. Discuss the various e-governance initiatives taken by government of Nepal and its various challenges.

E-governance refers to the use of information and communication technology (ICT) to enhance the efficiency, transparency, and accessibility of government services.

E-governance Initiatives in Nepal:

- **Nepal Government Portal**:

The official portal of the Government of Nepal (www.nepal.gov.np) serves as a central platform for accessing various government services and information online.

- **National ID Card Project**:

The government initiated a project to issue biometric national identity cards to citizens, aiming to improve identity authentication and streamline service delivery.

- **e-Government Procurement (e-GP)**:

The e-GP system facilitates the online procurement process, making it more transparent and efficient, and reducing corruption in public procurement.

- **Online Tax Filing and Revenue System**:

The government introduced an online platform for taxpayers to file tax returns and make payments, simplifying the tax compliance process.

- **e-Karyalaya**:

e-Karyalaya is an electronic document management system that allows government offices to manage their documents digitally, reducing paper usage and improving record-keeping.

- **e-Panchayat**:

e-Panchayat is an initiative to digitize local governance processes, enhancing the effectiveness of rural and municipal administrations.

- **Nagarik App**:

The Nagarik App provides a single platform for citizens to access various government services and information on their mobile devices.

Challenges in E-governance Implementation in Nepal:

- **Infrastructure and Connectivity**:

Limited internet infrastructure and connectivity in remote areas hinder the widespread adoption of e-governance services.

- **Digital Literacy**:

Low levels of digital literacy among citizens and government officials impede the effective use of e-governance platforms.

- **Language Barrier**:

Nepal is linguistically diverse, and providing services in multiple languages can be a challenge for e-governance initiatives.

- **Security and Privacy Concerns**:

Ensuring the security and privacy of citizens' data is a crucial challenge in e-governance implementation.

- **Interoperability**:

Lack of interoperability between different government systems can lead to inefficiencies and duplication of efforts.

- **Resistance to Change**:

Traditional bureaucratic practices and resistance to change can slow down the adoption of e-governance initiatives.

- **Funding and Sustainability**:

Adequate funding and long-term sustainability of e-governance projects are vital for their success.

- **Political Will and Administrative Support**:

E-governance requires strong political will and administrative support to overcome bureaucratic hurdles and implement reforms effectively.

Chapter 3: Computer Ethics and Ethical Theories

1. Explain 'Ethical Relativism' with example.

Ethical relativism is a philosophical theory that suggests moral principles and values are relative and context dependent. According to this view, there are no universal moral truths, and what is considered morally right or wrong varies from one culture, society, or individual to another.

Two types of Ethical Relativism:

a. **Cultural Relativism**

What is considered morally right or wrong varies from one culture to another.

To explain this concept with an example, let's consider the practice of eating meat:

In some cultures, like the United States and many Western countries, it is generally accepted that it is morally okay to eat meat. People raise animals like cows, chickens, and pigs for their meat, and it is a common part of their diet.

However, in certain other cultures or religions, like Hinduism or Jainism in India, the consumption of meat may be considered morally wrong or prohibited. In these societies, they might view animals as sacred beings or believe in the principle of non-violence, so they avoid eating meat to show respect for all living creatures.

b. **Individual Relativism**

What is considered morally right or wrong varies from one individual to another.

Example: Consider the issue of lying. According to individual relativism, whether lying is morally acceptable depends on an individual's perspective. For one person, lying may be justified in certain situations to protect someone's feelings, while for another person, lying might be seen as always wrong regardless of the circumstances. In this view, there is no objective truth about the morality of lying, only subjective opinions based on everyone's beliefs.

2. Utilitarian and Deontologist, though refer to different philosophies, may sometimes come to the same conclusion. Prove with an example.

Example: The Trolley (Train)Problem

The Trolley Problem is a classic thought experiment in ethics that presents a moral dilemma. Imagine you are standing next to a lever in control of a trolley (train) that is headed towards five people tied up on the tracks. You have two choices:

- Do nothing: If you do nothing, the trolley will continue its current track, and five people will be killed.
- Pull the lever: If you pull the lever, the trolley will switch to another track, where there is only one person tied up. In this case, one person will be killed, but five lives will be saved.

Let's analyse how utilitarianism and deontology might approach this situation:

- Utilitarian Perspective: Utilitarianism is the ethical theory that suggests we should act to maximize overall happiness or well-being. In the case of the Trolley Problem, a utilitarian might reason that pulling the lever is the morally correct action. By doing so, five lives are saved, which results in a greater net happiness or well-being compared to doing nothing and allowing five people to die.
- Deontological Perspective: Deontology is an ethical theory that focuses on the inherent rightness or wrongness of actions, regardless of their consequences. According to deontologists, certain actions are intrinsically right or wrong, regardless of the outcomes they produce. In the Trolley Problem, a deontologist might argue that pulling the lever is the morally right action because it adheres to the principle of "do not kill innocent people." By choosing to pull the lever, you are not directly intending to harm anyone; your intention is to save as many lives as possible.

In this specific example, both utilitarianism and deontology lead to the same conclusion: pulling the lever to divert the trolley to the track with one person is the morally right action.

3. Explain utilitarianism and Deontologist with example.

Example: The Organ Transplant Dilemma

Imagine you are a doctor working in a hospital where five patients are in critical condition, each requiring a different vital organ transplant to survive. One needs a heart, other needs lungs, a third needs a liver, and so on. Unfortunately, there are no available organs from deceased donors, and time is running out for these patients.

Then, a healthy person walks into the hospital for a routine check-up and happens to be a perfect match for all five patients. This individual is in excellent health and has no medical conditions. In a tragic twist, you discover that this person is also a volunteer for organ donation and has expressed their willingness to help others in need.

In this situation, utilitarianism would argue that the morally right action is to approach the healthy volunteer and ask for their consent to donate their organs. By doing so, the doctor is maximizing overall happiness and well-being. In this scenario, the greater number of lives saved, and the happiness generated for the patients, their families, and the healthy volunteer outweigh the potential risks involved in the organ transplant procedure.

Deontological says we should not do transplant from a healthy person. Baki afai lekha.

4. Explain Descriptive and Normative Claims with an example.

Descriptive Claims:

Descriptive claims are statements that describe or explain how things are or how they happen in the world. They are objective and factual statements based on observation, evidence, and empirical data. Descriptive claims do not express value judgments or prescribe how things should be; they simply provide information about the way things currently exist or have occurred.

Example of a Descriptive Claim:

"The Earth revolves around the Sun."

This statement is a descriptive claim because it describes a fact about the natural world based on scientific observations and evidence. It is an objective statement that can be verified through empirical research and does not carry any value judgment or normative implications.

Normative Claims:

Normative claims are statements that express judgments about how things ought to be or what actions are morally right or wrong. Unlike descriptive claims that focus on what is, normative claims focus on what should be. These claims involve values, ethics, and subjective beliefs about what is good, just, or proper.

Example of a Normative Claim:

"Everyone should have access to affordable healthcare."

This statement is a normative claim because it expresses an opinion about what should be the case in society. It reflects a value or moral judgment that affordable healthcare is a right that all individuals should enjoy. Unlike descriptive claims, normative claims cannot be proven true or false through empirical evidence alone, as they are based on subjective beliefs and values.

5. **What are the common legal and moral issues faced by software engineer. Explain with example.**

- Intellectual Property Rights:

Legal Issue: Software engineers must be careful to respect intellectual property rights, including copyright, trademarks, and patents. Creating software that breaks the law on the rights of others can lead to legal consequences.

Example: A software engineer develops a mobile application that closely resembles the interface and functionality of a popular app owned by another company. If the app is found to be a copy or derivative work of the original, the engineer may face legal action for copyright infringement.

- Privacy and Data Protection:

Legal Issue: Software engineers often work with sensitive user data, and they must follow privacy laws and data protection regulations to ensure the proper handling and safeguarding of personal information.

Example: A software engineer working on a social media platform but accidentally exposes user data, such as private messages or contact information, due to a security flaw in the system. This breach of data protection laws could lead to legal consequences for the engineer and the company.

- Ethical Use of Artificial Intelligence (AI) and Machine Learning:

Moral Issue: Software engineers involved in AI and machine learning development must deal with ethical considerations, such as bias, fairness, and the potential for harmful applications of AI technology.

Example: A team of software engineers develops an AI-powered hiring algorithm for a company. If the algorithm is found to discriminate against certain demographics, such as gender or race, it can raise ethical concerns about the fairness and impartiality of the hiring process.

- Software Bugs and Liability:

Legal Issue: Software engineers can face legal liability if their code contains critical bugs or vulnerabilities that result in financial losses, personal injuries, or property damage.

Example: A software engineer working on the code for a self-driving car inadvertently introduces a bug that causes the car's autonomous system to malfunction, leading to a serious accident. In this case, the engineer and the company could be held legally responsible for the consequences of the faulty software.

- Ethical Use of User Data:

Moral Issue: Software engineers often have access to vast amounts of user data, and they must consider ethical implications when using or sharing this data.

Example: A software engineer working for a social media platform has access to users' personal information and social interactions. Ethical concerns arise if the engineer misuses or sells this data without the users' knowledge or consent, violating their privacy rights.

6. Describe the computer ethics. Explain three different ways of organizing ethical issues.

1. Issue-based Approach:

i. Intellectual Property Rights:
ii. Privacy and Data Protection:
iii. Ethical Use of Artificial Intelligence (AI) and Machine Learning:
iv. Software Bugs and Liability:
v. Ethical Use of User Data:

2. Principle-Based Approach:

i. Utilitarianism:
ii. Deontology:

3. Stakeholder-Based Approach:

Chapter 4: PROFESSIONAL ETHICS

1. **Nepal Engineering Council Code of Ethics (2077 BS / 2020 AD)**:

- **Professional Competence**: Engineers should maintain a high level of professional competence and continuously update their skills and knowledge to provide quality engineering services.
- **Public Safety**: Engineers have a responsibility to ensure that their work does not compromise public safety, health, or the environment.
- **Integrity**: Engineers should act with honesty, integrity in all their professional dealings.
- **Confidentiality**: Engineers must protect the sensitive information acquired during their professional practice.
- **Conflict of Interest**: Engineers should avoid situations that could lead to conflicts of interest.
- **Sustainable Development**: Engineers should promote sustainable development, considering the long-term social, economic, and environmental impacts of their projects.
- **Ethical Use of Resources**: Engineers should use resources efficiently and avoid waste in their engineering projects.
- **Professional Development**: Engineers should actively engage in professional development activities to enhance their knowledge and skills.
- **Compliance with Laws and Regulations**: Engineers should comply with all relevant laws, regulations, and codes applicable to their professional practice.
- **Respect for Colleagues**: Engineers should treat their colleagues with respect and foster a collaborative and supportive work environment.

2. **IEEE Code of Ethics (**September 2021**)**

- **Preamble**: The Preamble introduces the code and emphasizes the importance of ethical behaviour in the IEEE community.
- **General Principles**:

a. **Responsibility to Public Health and Safety**: Members have a responsibility to ensure that their work does not compromise public safety, health, or the environment.

b. **Responsibility to Professional Integrity**: Members should act with honesty, integrity in all their professional dealings.

c. **Responsibility to Colleagues**: Members should treat their colleagues with respect and foster a collaborative and supportive work environment.

- **Application of the Code**: The Application of the Code section discusses how the principles of the IEEE Code of Ethics should be applied in various professional scenarios, providing guidance on ethical decision-making.
- **Commitment to Ethical Professional Conduct**: Members are encouraged to commit to ethical professional conduct and promote awareness and adherence

to the IEEE Code of Ethics within their organizations and professional communities.

ACM Code of Ethics:

General Moral Imperatives:

- **Contribute to Society:** Strive to make computing beneficial to society and use your skills for the greater good while minimizing harm to others.
- **Avoid Harm:** Avoid actions that could cause harm to others, including damage to property, privacy violations, and discriminatory practices.
- **Integrity:** Act with honesty, integrity in all their professional dealings.
- **Be Fair and Act Against Injustice:** Treat all individuals fairly and work against any unjust practices, particularly those related to computing.

More Specific Professional Responsibilities:

Intellectual Property: Respect copyrights and intellectual property rights, ensuring proper attribution and compliance with relevant laws and agreements.
Privacy and Security: Respect the privacy of others and protect personal data and sensitive information.
Accessibility: Strive to make computing resources and technologies accessible to all, without discrimination.
Respect for Others: Treat all individuals with respect and dignity, promoting diversity and inclusion.

Organizational Leadership Imperatives:

- **Promote Ethical Behaviour:** Encourage ethical conduct in the workplace and advocate for the adherence to ethical principles.
- **Positive Work Environment:** Create and support a work environment that fosters ethical behaviour, diversity, and professional growth.
- **Identify and Address Ethical Issues:** Recognize and address potential ethical issues, seeking resolution through appropriate channels.
- **Compliance with Policies:** Abide by all relevant laws, regulations, and organizational policies.

3. Define Whistleblowing.

Whistleblowing refers to the act of reporting information about unethical, illegal or wrongful activities within an organization or institution.

Types of whistleblowing:

- **Internal Whistleblowing**: Internal whistleblowing occurs when an employee or member of an organization reports wrongdoing or unethical behaviour to higher authorities or the management within the same organization.

  Example: An employee observes fraud financial practices within their company and reports it to their supervisor or the company's compliance officer.

- **External Whistleblowing**: External whistleblowing occurs when the individual reports the wrongdoing to individuals or entities outside the organization, such as government agencies, regulatory bodies, the media, or the public.
- Example: An employee at a pharmaceutical company becomes aware that the company is intentionally concealing adverse drug reactions and safety issues related to a recently released medication. Fearing that internal reporting might be suppressed, the employee decides to disclose this information to a government regulatory authority responsible for drug safety.
- **Anonymous Whistleblowing**: Anonymous whistleblowing occurs when the individual who reports the misconduct chooses to keep their identity secret. Example: An employee sends an anonymous letter to a regulatory agency detailing unethical practices in their workplace.
- **Open Whistleblowing**: Open whistleblowing is when the whistleblower openly comes forward and reveals their identity when reporting the wrongdoing. They take responsibility for their disclosure and are willing to be identified as the source of the information.

  Example: An employee holds a press conference or issues a public statement to expose corruption within their organization.

Conflict of Interest:

It is a set of circumstances that creates a risk that professional judgement or actions regarding a primary interest will be influenced by secondary interest.

Primary interest: Goal of profession (protection of client, health of client)

Secondary Interest: Includes personal benefit of employee or individual

What are the attributes of a profession? Verify that engineering is a profession, qualifying all these attributes.

Specialized Knowledge and Expertise: A profession requires specialized education, training, and knowledge in a specific field. Engineering certainly fulfils this attribute as engineers undergo extensive education and training to gain expertise in their chosen engineering discipline, such as civil engineering, mechanical engineering, or electrical engineering.

Ethical Standards and Code of Conduct: Professions must support ethical standards to ensure the well-being of society. Engineers are bound by codes of ethics, such as those established by professional organizations like the American Society of Civil Engineers (ASCE) or the Institution of Electrical Engineers (IEEE), which guide their professional conduct and emphasize integrity, honesty, and accountability.

Service Orientation: Professions have a service-oriented focus, aiming to benefit society or specific groups. Engineering is inherently service-oriented, as engineers design and create structures, systems, and technologies to address societal needs, improve infrastructure, and solve complex problems.

Ongoing Education and Development: Professionals engage in continuous learning to stay updated with the latest advancements in their field. Engineering requires ongoing education and professional development to keep up with evolving technologies, construction methods, and industry standards.

Professional Organizations and Licensing: Many professions have formal organizations or licensing bodies that establish standards for professional practice. Engineering is regulated in many countries, and obtaining a professional engineering license is often required to practice as a qualified engineer.

Hacker Ethics

Hackers' Ethics: The term "hacker" has evolved over time and can refer to different groups of individuals. It is essential to distinguish between ethical hackers (also known as "white-hat" hackers) and malicious hackers (also known as "black-hat" hackers). Ethical hackers adhere to a set of principles and ethics, while malicious hackers engage in unauthorized and harmful activities. Here are some key aspects of hackers' ethics:

- **Respect for Privacy:** Ethical hackers respect the privacy of individuals and organizations. They do not intrude on systems or networks without proper authorization.
- **Lawful Conduct:** Ethical hackers operate within the bounds of the law. They seek permission from system owners before testing or identifying vulnerabilities.
- **Professionalism:** Ethical hackers act in a professional manner and do not cause damage to systems or data during their assessments.
- **Disclosure and Responsible Reporting:** If ethical hackers discover vulnerabilities, they responsibly report them to the appropriate authorities or system owners, allowing for proper remediation.
- **Continuous Learning:** Ethical hackers continuously update their skills and knowledge to stay current with emerging threats and security measures.

- **Community Collaboration:** Ethical hackers may share their knowledge and expertise with the security community to enhance overall cybersecurity.

Netiquette

Netiquette, short for "Internet etiquette," refers to the set of guidelines and rules for respectful and courteous behaviour while interacting and communicating on the internet.

Respect Privacy: Avoid sharing personal information about yourself or others without permission. Protect your own privacy and be cautious about revealing sensitive information.

Be Respectful and Courteous: Treat others online with the same respect and courtesy you would in face-to-face interactions. Avoid offensive language, insults, or disrespectful behaviour.

Think Before Posting: Take a moment to consider the content and potential impact of your posts before sharing them. Once something is posted online, it may be challenging to retract or erase it completely.

Avoid Spamming and Trolling: Refrain from spamming others with excessive messages or engaging in trolling behaviours meant to provoke and upset others.

Observe Rules and Guidelines: Different online platforms may have specific rules and guidelines for participation. Familiarize yourself with the rules of each platform you use and follow them accordingly.

Respect Copyright and Intellectual Property: Give credit to the original creators of content when sharing or using their work. Avoid plagiarizing or using copyrighted material without permission.

Avoid Cyberbullying and Harassment: Treat others with kindness and refrain from engaging in cyberbullying or harassment. Report any inappropriate behaviour to platform administrators.

Chapter 8: Intellectual Property and Legal Issues

Secure Password Practices Issued by GoN (2067)

1. All system level password (root, enable, Network administration etc) must be changed on at least a quarterly basis. (barsha ko 4 choti password change garnu parxa)
2. All production system level password must be part of the Information security administered global password management database.
3. All user-level password (e.g.: email, web, desktop, computer) must be changed at least every two months. Recommended change interval is every month.
4. User account that has system level privileges granted through group membership or programs such as 'Sudo' must have a unique password from all other accounts held by that user.
5. Account lockout threshold = 4 failed login attempts. (4 choti failed password then lock the account)
6. Password history = Require several unique passwords before an old password may be used. This number should not be less than 24. (purano password feri use garna 24 ota unique password hale paxi matra painxa use garna)
7. Reset account lockout after
8. Account lockout duration is 30min to 2hrs.
9. Do not use 'Remember Password' option.
10. Do not access your organization information system from public network especially from cybercafe.

Electronics Transaction Act and Rules (2063):

1. **Electronic Records and Digital Signature**

a. Authenticity of Electronic Records:

i. A person can use a digital signature to authenticate electrical records.

ii. While transforming this record to other electrical record we must use asymmetric crypto system and hashing function.
iii. Person can verify the electronic record using a public key.

2. **Legal Recognition of Electrical Record:**
- Such authentication of electronic records shall have legal validity.
3. **Legal Recognition of Digital Signature:**
- As per this rule or act, a digital signature made accordingly shall also have legal validity.
4. **Electronic Records should be kept safely.**
5. **Electronic record may fulfil the requirement of submission of any original document**
6. **Secured Electronic Records**
- In relation to the matter of changing or not changing, if it has been verified by a recognized digital signature, such electronic records shall be deemed to be secured electronic records.
7. **Secured Digital Signature**
- The verified and recognized digital signature is considered as a secured digital signature.
8. **Offense Relating to Computer**
- To pirate, Destroy, Alter computer source code:

I. A fine of up to 2 lakhs and imprisonment for three years will be imposed.
- Unauthorized access in computer material.

I. A fine of up to 2 lakhs and imprisonment for three years will be imposed.
- Knowingly damage to any computer and information system.

i. A fine of up to 2 lakhs and imprisonment for three years will be imposed.
- Publication of illegal materials in electronic form.

i. A fine of up to 1 lakhs and imprisonment for five years will be imposed.
- To inform false statement.

i. A fine of up to1 lakhs and imprisonment for two years will be imposed.

Right to Information Act (2064):

1. **Flow of Information**

a. Right to Information
- According to this act, every Nepali citizen must have right to access the information held in public bodies.
- A public body cannot refuse to provide information without a reasonable cause.

b. Responsibility of public body
- Every public body must respect and protect the right of citizens to access information.

2. **Protection of Information**

- Public Body shall protect the information of personal nature held in for preventing unauthorized publication and broadcasting.
- It shall be a responsibility of an employee of a Public Body to provide information on any ongoing or probable corruption or irregularities or any deed taken as offence under the prevailing laws.

3. Commission

- There shall be an independent National Information Commission for the protection, promotion and practice of right to information.
- A Chief Information Commissioner and two other Information Commissioners shall be in the commission.

4. Punishment and Compensation

- If the Chief of a public Body or Information Officer, delay to provide information which must be provided on time without reason, shall be punished with a fine of Two Hundred Rupees per day so is delayed providing the information.

Copyright Act Nepal (2059):

1. Protection of a Copyright

- Copyright protection shall be extended to any work.
- Copyright protection will not be given to any work done on gender, religion, or discrimination.
- Registration is not compulsory.
- Economic right of work is given to the owner (author, coauthor).
- Copyright owners have right to reproduce, translate and revise the work.
- If anyone uses original work done by others, then they must mention the copyright owners name in their work.

2. Duration of Copyright

- The copyright protection lasts until owner's life and 50 years after death.
- If two or more people are involved, then it lasts until last person life and 50 years after last person death.

3. Circumstances where the copyright materials can be used without authorization

- It can be used for personal use by giving the citation.
- It can be used for research, education and learning purpose.

4. Transfer of copyright

- Copyright owner can give written agreement to others.
- But only the given rights can be used. If it is not mentioned in agreement, then such thing cannot be used.

5. **Infringement of Protected Rights and Punishment**

- A fine of up to 10,000 - 1 lakhs and imprisonment for 6 months will be imposed, if any people go against copyright laws.
- If it is repeated second time, then a fine of up to 20,000 - 2 lakhs and imprisonment for 1 year will be imposed.

Patent, Design and Trademark Act Nepal, (2022 B.S.)

Patent:

1. Acquisition of Patent right:
- A person desiring a patent must file an application according to this act.
- According to this act, once the person has obtained the patent then other people cannot allow to copy it or use it in their name.
2. Application for acquiring right over patent
- If person wants a patent in their name, then they must submit all the necessary evidence with the application along with their name, address, occupation, process of manufacturing, operating of their work, theory or formula in which patent is based.
3. Investigation by Department
- Once the department receives the application, then they start to investigate it.
4. Circumstances in which patent cannot be registered.
- If you failed to prove your work, then patent cannot be registered.
- If that work is already registered in another's name.
5. Duration of patent
- Up to 7 years from the registered date of patent.
6. A fine of up to 5 lakhs will be imposed if someone breaks the laws.

Design:

1. Acquisition of title to design:
- A person desiring a title to design must file an application according to this act.
- According to this act, once the person has obtained the design then other people cannot allow to copy it or use it in their name.
2. Application for acquiring right over design
- If person wants a design in their name, then they must submit all the necessary documents according to the act along with their design in the application.
- Investigation by Department
- Once the department receives the application, then they start to investigate it.
3. Duration of Design
- Up to 5 years from the registered date of design.
4. A fine of up to 50,000 will be imposed if someone breaks the laws.

Trademark:

Design:

1. Acquisition of title to trademark:

- A person desiring a title to trademark must file an application according to this act.
- According to this act, once the person has obtained the trademark then other people cannot allow to copy it or use it in their name.

2. Application for acquiring right over design

- If person wants a title to trademark in their name, then they must submit all the necessary documents according to the act along with their trademark in the application.

3. Investigation by Department

- Once the department receives the application, then they start to investigate it.

4. Duration of Trademark

- Up to 7 years from the registered date of design.

5. A fine of up to 1 lakh will be imposed if someone breaks the laws.

Chapter 7: Computer and Cyber Crimes

Phishing

Cybercriminals use fraud emails or websites to trick users into providing personal information, such as login credentials, credit card details, or other sensitive data. This is known as phishing.

Examples of Phishing:

- Email Phishing: Attackers send emails that appear to be from reputable sources, asking recipients to click on links or download attachments. These links may lead to fake websites designed to steal login credentials.

Example: A phishing email claiming to be from a well-known bank asks the recipient to click on a link to verify their account details due to a supposed security breach.

- Smishing (SMS Phishing): Attackers send text messages, asking recipients to click on links or download attachments. These links may lead to fake websites designed to steal login credentials.

Example: A text message claiming to be from a shipping company asks the recipient to click on a link to track a package, but the link leads to a fake website designed to steal personal information.

- Vishing (Voice Phishing): Attackers make phone calls pretending to be legitimate organizations or authorities, attempting to extract sensitive information over the phone.

Example: KBC bata phone gareko hajurle ek crore jitnu vayo vanera vanxa.. And voice from phone ko through bata money ultai lina khojxa asking them to give credit card details. Yei example lekha

Unauthorized access: Hacking, Cracking

Hacking is the act of gaining unauthorized access to computer systems, networks, or data with the intent to explore, manipulate, or exploit them.

Examples of Hacking:

- Password Cracking: A hacker uses various techniques, such as dictionary attacks, to guess or crack passwords and gain unauthorized access to user accounts.
- SQL Injection: In this type of attack, hackers insert malicious SQL code into a web application's input fields to manipulate the application's database, potentially gaining unauthorized access to sensitive data.
- Phishing Attacks: Hackers use deceptive emails, messages, or websites that appear legitimate to trick users into revealing their login credentials, allowing the hacker to gain unauthorized access to their accounts.
- Distributed Denial of Service (DDoS) Attacks: In a DDoS attack, hackers use multiple devices to flood a target system or network with excessive traffic, causing it to become unavailable to legitimate users.
- Ransomware: A type of malware that encrypts a victim's data, effectively locking them out of their own system until a ransom is paid to the hacker to provide the decryption key.

Denial of Service:

Denial of Service (DoS) is a type of cyber-attack where the attacker aims to disrupt the normal functioning of a computer system, network, or website by flooding it with a large volume of traffic or overwhelming it with excessive requests.

Malicious Program

Viruses

A Computer Virus is a type of computer program that attaches itself to other programs or files when executed and writes its own code so it can spread from one program to another, leaving infections as it travels. Much like human viruses, Computer Viruses can range in severity: some viruses cause only mildly annoying effects while others can damage your hardware, software or files. Almost all Computer Viruses are attached to an executable file, which means the virus may exist on your computer, but it cannot infect your computer unless you click on, run or open the malicious program.

Worm

Imagine a worm virus as a sneaky digital bug that can copy itself and spread from one computer to another without anyone realizing it. It doesn't need any help from you to move around. It travels through the internet and computer networks, looking for weaknesses to sneak into and infect other computers. Once inside a new computer, it starts making copies of itself and continues its journey to infect more computers. It can cause a lot of trouble by slowing down systems, eating up resources, or stealing information without the computer's owner even knowing it's there.

An example of a worm virus is the "ILOVEYOU" worm. When it infects a computer, it sends copies of itself to the person's contacts, making it spread very quickly. This caused a lot of chaos and disruptions when it first appeared because it spread so rapidly, affecting millions of computers worldwide.

Trojan horse

A Trojan horse is like a digital trickster hiding inside a harmless-looking gift. It pretends to be something useful or fun, like a game or an app, to tempt you into downloading and installing it. But once you open this gift and let the Trojan horse in, it secretly brings bad stuff along with it.

For example, imagine you download a new free game that claims to be super fun. You install it, and it seems to work fine, just like a regular game. But behind the scenes, the Trojan horse is up to no good. It might steal your private information, like passwords or credit card details, and send them to bad people. Or it could give hackers access to your computer, letting them control it without you knowing.

Computer Invasion of Privacy

Computer invasion of privacy refers to unauthorized access, collection, or monitoring of a person's private information or activities using computer technology.

- Unauthorized Access to Personal Accounts: A hacker uses various techniques, such as dictionary attacks, to guess or crack passwords and gain unauthorized access to user accounts.
- Spyware and Tracking Apps: Malicious software, such as spyware, can be installed on a person's computer or smartphone without their knowledge. These programs allow the attacker to monitor their online activities, messages, or location without consent.
- Data Collection by Websites and Apps: Some websites and mobile applications collect extensive user data without clear consents. This information may be used for targeted advertising or sold to third parties without explicit permission.
- Webcam and Microphone Hacking: Cybercriminals can remotely access a person's webcam or microphone without their knowledge, allowing them to spy on private conversations or activities.
- Location Tracking: Smartphone apps and devices with GPS capabilities can track a person's location, which could be exploited for malicious purposes if misused.

Online Harassment

Online harassment, also known as cyberbullying, is the act of using digital communication platforms to target, intimidate, or harm someone emotionally or psychologically.

Example of Online Harassment:

Imagine a teenager named Sarah who is active on social media. She becomes the target of online harassment by a classmate named Alex. Here's how the harassment unfolds:

- **Hurtful Messages:** Alex sends Sarah hateful and hurtful messages through direct messages on social media. The messages attack Sarah's appearance, intelligence, and personal interests, causing her to feel upset and humiliated.
- **Cyberbullying Posts:** Alex creates posts on their social media accounts, openly mocking and insulting Sarah. He encourages others to join in the harassment, leading to a barrage of hurtful comments on Sarah's posts.
- **Rumor Spreading:** Alex starts spreading false rumours about Sarah's personal life, alleging that she's involved in inappropriate relationships. These rumours cause Sarah's reputation and lead to gossip among her peers.
- **Fake Profiles:** To further torment Sarah, Alex creates fake profiles pretending to be other people and sends abusive messages to her and her friends. This leaves Sarah to unable to trust anyone online.
- **Cyberstalking:** Alex begins cyberstalking Sarah by tracking her online activities, monitoring her posts, and commenting on everything she shares. This constant monitoring makes Sarah feel like she has no privacy or escape from the harassment.
- **Online Isolation:** The harassment takes a toll on Sarah's mental well-being, leading her to withdraw from social media and online communities to avoid

further abuse. She starts feeling isolated and disconnected from her friends and peers.

Cyberstalking and Online Scams

Cyberstalking: Cyberstalking is a form of harassment where an individual uses electronic communication, such as social media, emails, or instant messaging, to repeatedly and maliciously pursue, monitor, or intimidate another person. It often involves unwanted attention, threats, and invasion of the victim's privacy.
Example: Jane and Mark recently broke up. Mark becomes obsessed with Jane and starts cyberstalking her. He constantly monitors her social media accounts, sends her threatening messages, and creates fake profiles to harass her. Mark also uses the information he gathers to target Jane's friends and family, creating a hostile and unsafe online environment for her.

Online Scams (same as phishing):
Online scams are deceptive schemes designed to trick people into providing personal information, money, or access to their devices. Scammers use various techniques to exploit their victims, often promising rewards or benefits that turn out to be fraudulent.
Example: A common online scam is the "phishing" scam. A scammer sends an email that appears to be from a legitimate bank or service provider, asking the recipient to verify their account information by clicking on a link. The link leads to a fake website that mimics the real one. Unsuspecting users provide their login credentials, unknowingly handing them over to the scammer. The scammer then gains unauthorized access to the victim's account, stealing their personal information or funds.

Cyber Terrorism

Cyberterrorism refers to the use of computer systems and networks to launch large-scale attacks on critical infrastructure, government systems, or public services with the intention of causing fear, disruption, or harm to society.

An example of cyberterrorism could be a coordinated cyber-attack on a country's power grid. Let's consider the fictional country of "Xylon":

- **Attack Preparation:** A group of cyberterrorists with advanced hacking skills identifies vulnerabilities in Xylon's power grid infrastructure. They gain access to critical control systems and conduct reconnaissance to understand the system's architecture.
- **Attack Execution:** On a specific date and time, the cyberterrorist group launches a sophisticated and simultaneous cyber-attack on multiple power stations and substations across Xylon. They use malware, denial-of-service attacks, and other hacking techniques to disrupt or disable critical control systems.
- **Power Outages:** As a result of the cyber-attack, several regions in Xylon experience widespread power outages. Hospitals, transportation networks, communication systems, and other essential services are severely affected, causing panic and chaos among the population.

- **Demands and Threats:** The cyberterrorist group takes responsibility for the attack through public statements online. They demand a ransom or make political or ideological demands, threatening to carry out more devastating cyber-attacks if their demands are not met.
- **Response and Investigation:** Xylon's government and law enforcement agencies launch an emergency response to mitigate the attack's impact and restore power. They also initiate a thorough investigation to identify the cyberterrorist group responsible for the attack.

Digital Forgery

Digital forgery, also known as digital manipulation, refers to the alteration, modification, or creation of digital content to deceive or mislead others.

Examples of Digital Forgery:

- **Image Manipulation:** A common form of digital forgery is manipulating images. For instance, someone might alter a photograph to remove or add objects or people, change facial expressions, or create composite images that combine elements from different photos. This can be done for humour, entertainment, or more malicious purposes to spread misinformation.
- **Deepfakes:** Deepfakes are a specific type of digital forgery that involves using artificial intelligence to superimpose one person's face onto another person's body in a video. Deepfakes can make it appear as if a person is saying or doing things they never actually did, raising concerns about the potential for spreading false information and damaging reputations.
- **Document Forgery:** In the digital realm, documents can be forged or manipulated to alter their content or create false records. For example, someone might alter a digital contract, financial statement, or identification document to deceive others for financial gain or to cover up illegal activities.
- **Audio Manipulation:** With advanced digital tools, it is possible to manipulate audio recordings to alter someone's voice or create fake audio clips that make it seem like a person said something they did not. This can be used to spread false information or for fraudulent purposes.
- **Video Editing:** Video content can be manipulated through editing software to distort the context or message of the original footage. This can be used to create misleading or fake news videos, potentially causing confusion or inciting unrest.

Chapter 6: Privacy

What are the best ways to protect an individual privacy in internet space?

- **Use Strong and Unique Passwords**: Create strong and unique passwords for online accounts. Avoid using common passwords.
- **Enable Two-Factor Authentication (2FA)**: Turn on 2FA whenever possible. This adds an extra layer of security by requiring a second form of verification, such as a code sent to your mobile device.
- **Review Privacy Settings**: Regularly review and update the privacy settings on social media platforms, apps, and websites. Limit the information shared with others to what is necessary.
- **Be Cautious with Personal Information**: Be cautious when sharing personal information online. Avoid providing unnecessary details on social media.
- **Use Virtual Private Networks (VPNs)**: Consider using a reputable VPN service, especially when accessing public Wi-Fi networks. VPNs encrypt internet traffic, enhancing online privacy and security.
- **Regularly Update Software and Apps**: Keep your operating system, software, and apps up to date. Updates often include security patches that protect against vulnerabilities.
- **Beware of Phishing Scams**: Be aware about phishing emails and messages that attempt to trick you into revealing personal information. Avoid clicking on suspicious links or providing sensitive data to unknown sources.
- **Clear Cookies and Cache**: Regularly clear cookies and browser cache to remove stored data that can track your online activities.
- **Protect Your Wi-Fi Network**: Secure your home Wi-Fi network with a strong password and encryption to prevent unauthorized access to your internet connection.

Privacy can still be maintained even when the individuals are exposed to social networking sites. Do you agree or disagree? Justify your opinion.

- **Privacy Settings**: Social networking sites typically offer privacy settings that allow users to control who can see their posts, profile information, and personal details. By adjusting these settings, users can limit the visibility of their content to a select group of friends or followers, enhancing their privacy.
- **Selective Sharing**: Users have the freedom to be selective about what they share on social media. They can choose not to post sensitive or personal information, ensuring that only necessary and non-sensitive details are accessible to others.

- **Limited Personal Data**: Social media platforms usually require certain personal information during account creation, but users can often choose to provide minimal data and avoid disclosing more sensitive details.
- **Review and Removal**: Users can review their past posts, comments, and interactions on social media and remove any content that compromises their privacy or is no longer relevant.
- **Blocking and Unfriending**: Social media platforms offer options to block or unfriend individuals, allowing users to control who can interact with them and view their content.

Is it possible to enforce censorship in cyberspace?
Enforcing censorship in cyberspace is a challenging and complex task. While it is possible to implement certain censorship measures, achieving complete control over the vast and dynamic online environment is difficult due to several factors:

- **Global Nature of the Internet**: The internet operates on a global scale, crossing national borders and jurisdictions. Different countries have diverse laws and regulations regarding censorship, making it difficult to enforce consistent measures worldwide.
- **Technological Advancements**: As technology evolves, individuals and organizations find ways to bypass censorship measures. Tools like virtual private networks (VPNs) and encrypted communication platforms allow users to access blocked content and communicate privately.
- **Privacy Concerns**: Censorship often involves monitoring and surveillance, raising privacy concerns among internet users. Striking a balance between censorship and protecting individual freedoms can be a delicate challenge.
- **Cat-and-Mouse Game**: As censorship is enforced, some individuals and groups may attempt to find loopholes or use creative methods to evade restrictions, leading to a constant cat-and-mouse game between censors and those seeking to bypass them.
- **Censorship Evasion**: Internet users can use mirror sites, proxy servers, or decentralized platforms to circumvent censorship attempts and distribute content freely.
- **Over blocking and Collateral Damage**: Enforcing censorship can lead to over blocking, where legitimate content is also restricted inadvertently. This may impact freedom of expression and access to information.
- **Global Content Distribution**: Digital content can be hosted on servers located in different countries, making it challenging for a single jurisdiction to enforce censorship effectively.

1. What are offensive speech and censorship? Explain the importance of censorship by illustrating and example.

Offensive Speech:

Offensive speech refers to expressions, statements, or content that can be considered rude, disrespectful, harmful, or insulting to individuals or groups based on their race, religion, ethnicity, gender or other characteristics.

Censorship:

Censorship involves the suppression or restriction of information, speech, or media content deemed harmful, obscene, or inappropriate by authorities or governing bodies.

Importance of Censorship:

- **Protecting Public Safety**: Censorship helps prevent the spread of harmful or dangerous information that could lead to physical harm, violence, or illegal activities.
- **Safeguarding Vulnerable Populations**: Censorship shields children, minors, and individuals in vulnerable situations from exposure to inappropriate or harmful content.
- **Maintaining Social Order**: Censorship prevents the dissemination (spreading something) of hate speech, incitement to violence, and misinformation that could lead to social unrest and conflicts.
- **Preserving Public Morals**: Censorship ensures that content aligns with societal norms and values, preventing the spread of offensive or morally objectionable material.
- **Promoting Responsible Media Use**: Censorship encourages responsible use of media platforms, discouraging the dissemination of false or harmful information.
- **Preventing Cybersecurity Threats**: Censorship can help block malicious content, such as phishing scams or malware, protecting users from cyber threats.

Example of Censorship:

Let's consider an example of social media censorship. A user on a popular social platform starts promoting harmful and violent ideologies, encouraging others to engage in illegal activities. This content can have a negative impact on impressionable individuals and may lead to real-world violence or unlawful behaviour.

To protect public safety and maintain a safe online environment, the social media platform employs censorship measures. The offending content is promptly identified and removed, preventing its further spread. By taking such action, the platform helps ensure that the harmful ideologies do not influence others and that the platform remains a place for positive interactions and responsible use.

Privacy of Consumer Information:

Databases and Personal Records:

Data Security Measures: Organizations must implement data security measures to protect consumer information from unauthorized access. This includes encryption of data, firewalls etc.

Consent and Transparency: Consumers should be informed about what data is being collected, why it is being collected, and how it will be used.

Limited Data Collection: Organizations should only collect the minimum amount of data necessary to fulfil their business purposes. Collecting excessive or unnecessary information increases the risk of data breaches.

Data Storage and Retention: Consumer data should be stored securely and only retained for as long as necessary.

Employee Training: Organizations should provide regular training to employees on data privacy best practices and the importance of handling consumer information responsibly.

Email Privacy

Email privacy refers to the protection of your personal information and messages from being accessed or read by unauthorized individuals. When you send an email, it travels through various servers and networks before reaching the recipient. During this journey, there's a risk that someone could intercept or view your email without your permission.

An email without proper privacy measures is like an open letter. Without encryption and security measures, hackers or unauthorized people could potentially read your email while it travels across the internet.

To ensure email privacy, you can use tools like encrypted email services, which seal your messages in a digital envelope so that only the intended recipient can decrypt and read them.

Web Privacy

Web privacy refers to the protection of your personal information and online activities from being accessed or read by unauthorized individuals.

- **Cookies and Tracking:** When you visit websites, they may place small files called cookies on your device. These cookies can track your browsing behaviour, preferences, and activities across different sites. While some cookies are necessary for website functionality, others are used for targeted advertising or user profiling, which can compromise your privacy.
- **Social Media and Data Collection:** Social media platforms collect a vast amount of personal information about their users, including your interests, location, and connections. This data can be used to create detailed profiles and targeted ads, potentially revealing more about you than you intend to share.
- **Search Engines and Personalized Results:** Search engines like Google store your search history and use it to provide personalized search results and ads. While this can improve user experience, it also means your search behaviour is being tracked and analysed.
- **Online Shopping and Payment Information:** When making purchases online, you need to provide payment information. Ensuring that the website has secure encryption (https://) is crucial to protect your financial details from potential hackers.

To protect your web privacy, you can take the following steps:

- **Use Virtual Private Networks (VPNs)**: Consider using a reputable VPN service, especially when accessing public Wi-Fi networks. VPNs encrypt internet traffic, enhancing online privacy and security.
- **Review Privacy Settings**: Regularly review and update the privacy settings on social media platforms, apps, and websites. Limit the information shared with others to what is necessary.
- **Clear Cookies and Cache**: Regularly clear cookies and browser cache to remove stored data that can track your online activities.
- **Use Private Browsing Mode:** Enable private browsing or incognito mode in your web browser to prevent it from saving your browsing history and cookies.
- **Be Cautious with Personal Information:** Be mindful of the information you share online, especially on social media, as it can be used to build a comprehensive profile of you.

Chapter 5: Risk and Responsibilities

How would you manage the risk of complex software system in a controlled manner?

1. Risk Identification:

- Identify potential risks related to the software system. This may include security vulnerabilities, performance bottlenecks etc.

2. Risk Assessment:

- Evaluate the impact of each identified risk. Rank risks based on those with the highest potential impact.

3. Risk Mitigation Planning:

- Develop a risk mitigation plan for each high-priority risk. The plan should outline specific actions to reduce the impact of the risk.

4. Architecture and Design Considerations:

- Design the software system with modularity, clear interfaces, and well-defined responsibilities for each component. A well-architected system can better manage complexity and reduce the likelihood of errors.

5. Code Quality and Review:

- Conduct code reviews to identify and fix potential issues early in the development process.

6. Automated Testing:

- Implement comprehensive automated testing, including unit tests, integration tests, and end-to-end tests. This helps identify and resolve defects before they become critical issues.

7. Security Measures:

- Implement security best practices, including encryption, access controls, input validation, and secure authentication mechanisms, to protect against potential security vulnerabilities.

8. Backup:

- Regularly backup data to ensure business continuity in case of unexpected failures or data loss.

10. Monitoring and Logging:

- Implement monitoring and logging mechanisms to track the system's performance and identify potential issues in real-time.

What do you mean by complex network? Which characteristics are to be considered for finding complexity?

A complex network, often referred to as a complex system or a complex network system, is a network with a large number of interconnected elements or nodes, where these nodes interact exhibit emergent properties that cannot be easily understood by simply examining the individual elements in isolation.
Characteristics to consider when identifying complexity in a network include:

- **Large Number of Nodes**: Complex networks typically have a large number of nodes (elements) that are interconnected in complicated ways. The sheer number of nodes can contribute to the complexity of the network.
- **Connectivity**: The way nodes are connected to each other is crucial in determining complexity. Networks with dense and intricate connections tend to be more complex than those with simple and sparse connections.
- **Clustering Coefficient**: Clustering coefficient measures the tendency of nodes in a network to form clusters or groups. Higher clustering indicates more complex network system.
- **Short Average Path Length**: The average shortest path length between any two nodes in the network can also indicate complexity. Networks with short average path lengths tend to facilitate efficient information flow.
- **Assortativity**: Assortativity measures the tendency of nodes with similar degrees to be connected to each other. Networks can be assortative (similar degrees connected) or disassortative (dissimilar degrees connected).

Define software risks? Explain different software risk with suitable example.

They are uncertainty things that may occur is software and leads to heavy losses.

Why are software entities considered complex? What are the methods of resolving complexity of a software? Can these complexities be resolved completely? Give your view.

Software entities are considered complex due to several factors:

- **Size and Interconnectedness**: Software systems can be large, consisting of thousands or millions of lines of code, with numerous interconnected modules and components. Managing and understanding such intricate relationships can be challenging.
- **Abstraction and Hierarchy**: Software is often designed using abstraction and hierarchical structures, which can lead to deep layers of functionality. Understanding how these layers interact and ensuring they work seamlessly can be complex.
- **Dynamic Behaviour**: Software is not static; it interacts with users, external systems, and the environment. Dealing with changing conditions and dynamic behaviour adds complexity to software development and maintenance.

- **Requirements and Changes**: As requirements evolve and change, so does the software. Managing these changes while preserving the integrity of the system can be complex.
- **Technical Diversity**: Modern software often integrates various technologies, libraries, and platforms, making it more challenging to proper functioning.

Methods of Resolving Complexity of Software:

- **Modularization**: Divide the software into smaller, manageable modules with well-defined interfaces. This approach allows developers to focus on individual pieces of functionality, making it easier to understand, test, and maintain.
- **Abstraction and Encapsulation**: Use abstraction to hide complex implementation details behind simpler interfaces. Encapsulation ensures that internal details are not exposed, reducing the burden on other parts of the system.
- **Object-Oriented Design**: Object-oriented programming allows developers to model real-world entities as object, reducing complexity.
- **Code Reviewing**: Regularly review code to improve its structure and remove redundant or unnecessary complexity.
- **Automated Testing**: Implement comprehensive testing methodologies to catch issues early and ensure that the software behaves as expected.
- **Documentation**: Maintain clear and up-to-date documentation, including code comments, architectural diagrams, and user guides, to aid understanding and future maintenance.

Can These Complexities be Resolved Completely?

While various techniques can help manage and reduce complexity, it is challenging to eliminate it entirely, especially in large and sophisticated software systems. As a software project evolves and incorporates new requirements or technologies, new complexities may emerge.

However, by employing sound software engineering practices, developers can minimize and control complexity, making the software more maintainable, extensible, and comprehensible.